GEOAI 04 change report: Pune city edge (Hinjewadi)

Totals

Raw detections: 6830
Real changes after filters: 475
Area changed: 141.7 ha
Cropland affected: 8.67 ha (upper bound)
Built-up land affected: 31.76 ha
critical: 4
high: 39
medium: 123
low: 309

Open this result: /scene/pune-hinjewadi

Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	critical	built_up_gain	9.46	0.908	new construction / built-up: vegetation index fell (-0.14), water index rose (+0
2	critical	built_up_gain	4.16	0.959	new construction / built-up: built-up index rose (+0.16), SWIR/blue ratio (paved
3	critical	built_up_gain	4.94	0.924	new construction / built-up: built-up index fell (-0.12), vegetation index fell
4	critical	built_up_gain	2.325	0.981	new construction / built-up: built-up index rose (+0.18), water index rose (+0.0
5	high	water_loss	2.205	0.981	water receded: vegetation index rose (+0.26), water index fell (-0.46), SWIR/blu
6	high	built_up_gain	1.665	0.971	new construction / built-up: built-up index rose (+0.15), vegetation index fell
7	high	built_up_gain	2.165	0.898	new construction / built-up: vegetation index fell (-0.15), water index rose (+0
8	high	built_up_gain	1.28	0.982	new construction / built-up: water index rose (+0.05)
9	high	built_up_gain	1.415	0.958	new construction / built-up: vegetation index fell (-0.12), water index rose (+0
10	high	built_up_gain	1.42	0.951	new construction / built-up: vegetation index fell (-0.14), water index rose (+0

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero